Northeastern University
College of Engineering
Department of Electrical and Computer Engineering

EECE 2140: COMPUTING FUNDAMENTALS FOR ENGINEERS
Spring 2026
Course Syllabus

GENERAL COURSE INFORMATION

This course provides ECE students with a solid foundation in C++ programming principles, emphasizing practical applications relevant to hardware interfaces, engineering applications, system design, and embedded systems. Students will develop skills to implement software-engineered programming implementations, while leveraging current state-of-the-art tools that can improve code quality and coding efficiency. Students will explore the use of AI in program development.

NUMBER OF CREDIT HOURS: 4 Credit Hours.

INSTRUCTOR AND GTA INFORMATION

- **Dr. Fatema Nafa**
- **Office**: **426 Dana Research Center, Boston, MA 02115**
- **Telephone**: +1 (617) 373 -XXXX | E-mail: f.nafa@northeastern.edu
- **Office Hours:** **Wednesday and Fridays from 10:00 AM to 11:30 AM**. or by appointment.
  - ✓ Please send me an email if you would like to meet at some other time.
  - ✓ During the office hours, I can meet with you either in-person or virtually through Zoom or Microsoft Teams.
  - ✓ Please note that the office hours may vary from week to week. Check the announcements posted on the course website on Canvas.
- **Graduate Teaching Assistant (GTA)**
- **Office Hours:**
- **Zoom Meeting Link for Office Hours:**

CLASS MEETING TIME AND LOCATION

- **Section 03**: **Wednesday and Fridays 11:45 am - 1:25 pm in Hayden Hall 009.**

Note: Throughout the semester, if and whenever Northeastern University moves to the remote learning format, all class and lab sessions that need to be held remotely will be synchronous through the Zoom application:

CO-REQUISITE: None.

Course Details

Prerequisites: none

Required Materials: Textbook: "C++ Primer" by Stanley B. Lippman, Josée Lajoie, and Barbara E. Moo (5th Edition)

Required Software

- **Visual Studio Code** (VS Code): A lightweight, flexible code editor ideal for C++ development. Supports extensions like C/C++ IntelliSense, debugging, Git, and GitHub Copilot.

- **C++ Compiler:** One of the following is required: **GCC** (GNU Compiler Collection) Common on Linux.
- **CMake:** A cross-platform build system used to configure and compile multi-file C++ projects.
- **GitHub:** A platform for version control and collaboration. Students will use GitHub to manage repositories, track changes, and submit assignments.
- **GitHub Copilot:** An AI-powered code assistant that suggests code completions, documentation, and tests in real time. Integrated into VS Code, it helps accelerate learning and supports writing cleaner, more efficient C++ code.
- **Overleaf:** An online LaTeX editor useful for creating technical reports, lab documentation, and project write-ups.

COURSE OBJECTIVES AND LEARNING OUTCOMES

By the end of this course, students will be able to:

1. Write, compile, and debug C++ programs
2. Develop/run code on Linux-based systems
3. Apply object-oriented programming principles
4. Implement data structures and simple algorithms in C++
5. Create modular, reusable, and maintainable code
6. Use IDEs for code development, and leverage AI tools for coding assistance
7. Leverage AI tools to support code development and debugging

Teaching Strategy

Students will be working on 2 assignments per week. One will be a lab-based assignment. The second is a mini-project that is completed outside the class.

COURSE GRADING

TENTATIVE GRADING SCALE (PERCENTAGE %):

Letter Grade	A	A-	B+	B	B-	C+	C	C-	D+	D	D-	F
% Score	≥ 94	≥ 90	≥ 87	≥ 84	≥ 80	≥ 77	≥ 74	≥ 70	≥ 67	≥ 64	≥ 60	<60

Assessment Component	Description	% of Grade
Mini-Project Assignments	3-4 structured mini projects applying core concepts (individual or pairs)	**20%**
In Class Work	Weekly or bi-weekly in-class coding labs and participation	**10%**
Midterm Exam	Coding and conceptual understanding (hands-on)	**25%**
Course Project (Final project)	Multi-week cumulative project with proposal, checkpoint, demo, and report	**30%**
Quizzes	Short assessments covering weekly or bi-weekly material	**10%**
Reading & Engagement	Assigned readings, reflections, prep questions, participation	**5%**
Total		**100%**

Note: At the instructor's sole discretion, the grades may be curved, the above decision boundaries may be adjusted in your favor, or the above tentative weights may be revised, but you are guaranteed at least the letter grade shown here if you obtain the corresponding score.

IMPORTANT COURSE POLICIES

This course encompasses a comprehensive structure comprising lectures, homework assignments, quizzes, examinations, and Final Projects.

- **Lecture Arrangements:** The course lectures will be conducted fully in person adhering to the university's predetermined schedule. However, should any modifications arise in the class modularity, it will be the instructor's responsibility to promptly communicate these changes through Canvas, our primary online learning platform. Possible adjustments to class timings, content, or structure may occur due to various factors. Remaining well-informed is crucial to ensure a smooth learning journey.
- **Effective Communication:** To ensure seamless communication, please inform the instructor promptly if you encounter any difficulties accessing Canvas or if you have not received the welcome email I sent before the first day of class. Kindly address this within the first week of the course.
- **Canvas Notifications:** All notifications regarding alterations to the class modularity will be transmitted through your registered email address on Canvas. Consistently monitoring your email and staying updated with Canvas notifications will enable you to stay abreast of any changes or updates related to the course. Your attentiveness to these notifications is genuinely appreciated and will significantly contribute to your success in the course.
- **Lectures and hands-on:** Lectures, conducted in person, will take place twice a week. A corresponding hands-on session is associated with each lecture. The hands-on sessions will complement the lecture content and facilitate the practical application of the materials covered. Attendance and active participation in both lectures and hands-on are obligatory. Following the lectures, hands-on will be conducted to reinforce the learning process. The concluding 20 minutes of each lecture will be dedicated to a review of the lecture content. These hands-on sessions will offer immediate opportunities to apply the concepts discussed and serve as a preliminary introduction to subsequent programming assignments.
- **Written Assignments:** Every week it concludes with a problem set or a programming assignment. These assignments are to be individually undertaken by the students.
- **Quizzes**: Quizzes are scheduled during lecture times and are intended for individual completion by students, and an announcement will be sent in advance via Canvas platform.
- **Examinations**: Examinations will be administered in class. The assessment structure comprises two examinations. It is advised not to make any conflicting arrangements on the designated exam date.
- **Projects**: As a vital component of this course, a final project serves as a significant assessment. The project can be a collaborative effort, accommodating teams consisting of a maximum of two individuals per group. You have the liberty to choose your project partner. However, both the submission of the final project and the presentation should occur only after reaching a mutual agreement on the collaborative work between the students. These guidelines and procedures provide the framework for an effective and structured learning experience throughout the course.
- **Attendance and Participation: Attendance and active participation in every lecture are mandatory to foster an effective learning environment. Students are expected to be punctual and engaged.**

- **Academic Responsibility: Students** are accountable for all class materials, exams, and announcements. No excuses will be considered for incomplete assignments. Responsibility for obtaining missed class notes rests with the student, as the instructor does not provide slides or notes.
- **Assignment Submissions:** All assignments must be submitted on time via the online Canvas system. Late submissions are not accepted unless they are due to family or medical emergencies. Files can be resubmitted multiple times before the deadline. Any submission past the deadline is marked as late. Email responses are generally within 24 hours on weekdays.
- **ChatGPT:** AI-based tools such as ChatGPT can be valuable for improving your learning for example, asking for clarifications, generating examples, or receiving tutoring-style explanations of concepts. However, these tools must not be used to produce your final answers or complete assignments on your behalf. The work you submit must reflect your own understanding, reasoning, and coding. Assignment instructions, due dates, and submission links will be on Canvas. Dishonesty or cheating is strictly prohibited. While open discussions and collaborative learning are encouraged, the final submitted work must always be your own. Late submissions will incur penalties.

- **Study Groups and Extra Credit:** Study groups are welcomed, but individual answers should be unique. Extra credit can be added to assignments, exams, or lab work to uplift class averages. In **case of disputes, extra credit may be revoked.**
- **Exam Policies: The instructor may modify the exam format to accommodate student needs when necessary. Any changes will be communicated through email. It is the student's responsibility to contact the instructor in advance to arrange a make-up exam.**

Student Responsibilities or Tips for Success in the Course

This course is a rewarding yet intensive journey into programming, where challenges will increase in complexity as the semester progresses. Our goal is not just to learn how to write code but how to craft efficient, elegant, and thoughtful solutions to real-world problems.

To succeed in this course, keep the following in mind:

Ask for Help Often!

You're not expected to know everything right away. I'm here to support your learning, and your questions help everyone in the class. Please ask during lectures, labs, or office hours — your curiosity drives deeper understanding for all.

AI is Your Learning Partner Not Your Substitute

You are encouraged to use tools like ChatGPT, Copilot, or other AI assistants to explore, debug, and learn.

Use them to:

- Understand tricky error messages
- Review code style or syntax
- Experiment with small examples

But remember: the goal is for you to learn how to solve problems. Submitting AI-generated work without understanding will hurt you in the long run — and in later, harder courses.

Be Patient with Bugs

Programming can be frustrating — and that's okay. Everyone gets stuck. Take breaks, talk to classmates, or email me. Spending time on problems is part of learning, and each bug you fix makes you stronger.

Test Early, Test Often

Run your code after every few lines. It's much easier to debug small chunks than entire files. Use print statements, comments, or test inputs to verify your progress as you go.

Back It Up

Save your work frequently and use cloud tools like Google Drive, GitHub, or OneDrive. Accidental file loss is no fun but it's preventable.

Collaboration is Good —Copying is Not

Talk to peers, brainstorm together, and help each other grow. But always submit your own work. Plagiarism is not acceptable, understanding your code is the foundation of your success.

Final Tip: Trust the Process

This course will stretch your problem-solving muscles. Stick with it, practice consistently, and use all available resources including your classmates, your instructor, and smart tools the right way.

Week / Topic	Learning Outcomes	Assignments & Activities
Week 1: Introduction to C++ and IDEs • C++ overview and history • Linux overview • Comparison with other languages • Linux terminal basics • Git fundamentals (init, add, commit, push, pull)	Identify the role of C++ in engineering applications.Explain Linux-based development workflows.Use Git and GitHub for version control.	**Reading:** Syllabus + Intro to C++ **Lab 01:** Set up GitHub repo, VS Code, Makefile on Linux
Week 2: C++ Fundamentals • Variables and data types • Operators and expressions • Control structures • Functions and debugging	Write and debug basic C++ programs.Apply control structures and functions correctly.	**Lab 02:** Debug faulty program + decoding task **Mini Project 01:** Terminal-based calculator (GitHub submission)
Week 3: Copilot & Debugging Workflow • GitHub Copilot setup • AI-assisted coding • Debugging with VS Code • Best practices for AI usage	Use AI tools responsibly for code generation and debugging.Evaluate AI-assisted vs manual solutions.	**Lab 03:** Copilot vs manual debugging comparison **Mini Project 02:** Copilot-powered sorting (numbers & words)
Week 4: Arrays, Pointers, and References • Arrays and memory layout • Pointers and dereferencing • References • Dynamic memory allocation	Apply pointer-based memory addressing.Differentiate pass-by-value vs pass-by-reference.	**Lab 04:** Matrix algebra using pointers **Mini Project 03:** Image denoising using matrices
Week 5: Functions and Program Structure • Function overloading • Inline and default functions	Design modular, multi-file C++ programs.Organize projects using best practices.	**Lab 05:** Modular program with function overloading **Mini Project 04:** Queue system using header/source files

Week / Topic	Learning Outcomes	Assignments & Activities
• Namespaces • Header/source files • Command-line arguments		
Week 6: Object-Oriented Programming (Basics) • Classes and objects • Constructors/destructors • Encapsulation	Design and implement simple C++ classes using OOP principles.	**Lab 06:** Class design and implementation **Mini Project 05:** Circuit simulator – base components
Week 7: Object-Oriented Programming (Advanced) • Inheritance and polymorphism • Virtual functions • Abstract classes	Apply inheritance and polymorphism in engineering models.	**Lab 07:** Class hierarchy implementation **Mini Project 06:** Circuit simulation with AI visualization
Week 8: Memory Management • Stack vs heap • new/delete • Smart pointers • Memory leaks	Manage memory safely using modern C++ techniques.	**Lab 08:** Memory management tools **Mini Project 07:** Graph routing engine (manual + smart pointers)
Week 9: Midterm & Project Introduction • Midterm exam • Final project options • Visualization requirement	Demonstrate mastery of core C++ concepts.Plan a complete engineering project.	**Midterm Exam** **Lab 09:** Team setup & project planning**Project Proposal Due**
Week 10: File I/O and Streams • Text and binary files • Stream manipulators • Error handling	Implement robust file-based applications.	**Lab 10:** Data logger with text search**Final Project – Iteration 01:** Repo setup & README
Week 11: STL Containers • vector, list, deque • map, set • stack, queue	Use STL containers to implement efficient data structures.	**Lab 11:** STL container implementations**Final Project – Iteration 02:** Core implementation + pseudocode
Week 12: STL Algorithms & Iterators • Algorithms • Iterators • Lambdas • Complexity	Apply STL algorithms to engineering problems.	**Lab 12:** STL algorithm problem solving**Final Project – Iteration 03:** Exception handling + code review
Week 13: Exception Handling & Reflection • try-catch • Custom exceptions • AI reflection	Implement robust error handling.Reflect on ethical and effective AI usage.	**Lab 13:** Exception handling**Reflection:** AI-assisted programming write-up
Week 14: Modern C++ & Integration • Design patterns• Modern C++ features • Git merging	Integrate advanced C++ features in large projects.	**Lab 14:** Advanced Copilot features**Final Project:** Peer review & Git merge

Week / Topic	Learning Outcomes	Assignments & Activities
Week 15: Final Project Presentations • Project demos • Course wrap-up	Communicate technical solutions clearly.Evaluate peer work.	**Lab:** Final project showcase & peer review

IMPORTANT UNIVERSITY POLICIES

ACADEMIC INTEGRITY, PLAGIARISM, DISHONESTY, AND CHEATING POLICY: A commitment to the principles of academic integrity is essential to the mission of Northeastern University. The promotion of independent and original scholarship ensures that students derive the most from their educational experience and their pursuit of knowledge. Academic dishonesty violates the most fundamental values of an intellectual community and undermines the achievements of the entire University.

As members of the academic community, students must become familiar with their rights and responsibilities. In each course, they are responsible for knowing the requirements and restrictions regarding research and writing, examinations of whatever kind, collaborative work, the use of study aids, the appropriateness of assistance, and other issues. Students are responsible for learning the conventions of documentation and acknowledgment of sources in their fields. Northeastern University expects students to complete all examinations, tests, papers, creative projects, and assignments of any kind according to the highest ethical standards, as set forth either explicitly or implicitly in this Code or by the direction of instructors. The University is committed to investigating any allegation of violations of academic integrity against a student. Violations include, but are not limited to, plagiarism, cheating, fabrication, unauthorized collaboration, and academic misconduct. Sanctions for violations of academic integrity are administered through the Office of Student Conduct and Conflict Resolution (OSCCR) in conjunction with other University offices as deemed appropriate. It is generally the responsibility of the faculty member overseeing the academic activity to report the violation to OSCCR and to determine the appropriate sanction. A student who believes they have been wrongly sanctioned has a right to an appeals process. Go to http://www.northeastern.edu/osccr/academic-integrity-policy/ to access the full academic integrity policy.

STUDENT ACCOMMODATIONS: Northeastern University and the Disability Resource Center (DRC) are committed to providing disability services that enable students who qualify under Section 504 of the Rehabilitation Act and the Americans with Disabilities Act Amendments Act (ADAAA) to participate fully in the activities of the university. To receive accommodations through the DRC, students must provide appropriate documentation that demonstrates a current substantially limiting disability. For more information, visit http://www.northeastern.edu/drc/getting-started-with-the-drc/.

DIVERSITY AND INCLUSION: Northeastern University is committed to equal opportunity, affirmative action, diversity, and social justice while building a climate of inclusion on and beyond campus. In the classroom, members of the University community work to cultivate an inclusive environment that denounces discrimination through innovation, collaboration and an awareness of global perspectives on social justice. It is my intention that students from all backgrounds and perspectives will be well served by this course, and that the diversity that students bring to this class will be viewed as an asset. I welcome individuals of all ages, backgrounds, beliefs, ethnicities, genders, gender identities, gender expressions, national origins, religious affiliations, sexual orientations, socioeconomic background, family education level, ability, and other visible and nonvisible differences. All members of this class are expected to contribute to a respectful, welcoming, and inclusive environment for every other member of the class. Your suggestions are

encouraged and appreciated. Please visit http://www.northeastern.edu/oidi/ for complete information on Diversity and Inclusion.

TITLE IX: Title IX of the Education Amendments of 1972 protects individuals from sex or gender-based discrimination, including discrimination based on gender-identity, in educational programs and activities that receive federal financial assistance. Northeastern's Title IX Policy prohibits Prohibited Offenses, which are defined as sexual harassment, sexual assault, relationship or domestic violence, and stalking. The Title IX Policy applies to the entire community, including male, female, transgender students, faculty, and staff. In case of an emergency, please call YJJ. Please visit www.northeastern.edu/titleix for a complete list of reporting options and resources both on- and off-campus.

INSTRUCTIONAL CONTINUITY AND CLASS CANCELLATION: Instructional continuity refers to the continuation of instruction during unforeseen campus closure or instructor absence. Should the need to cancel a class session occur, students will be notified through Canvas and/or their Northeastern email address. Students are responsible for all course material provided through this instructional continuity plan. The following procedures will be in place to ensure continuity of instruction in this course:

- The lecture scheduled to be delivered in a cancelled class session will be recorded either synchronously or asynchronously and posted on Canvas for the students to watch within Db hours of the cancelled class session. In lieu of the recorded lecture, students might be asked to watch one or multiple videos from reputable online resources. A reading or an online problem-solving practice session with the instructor or teaching assistant for the course might also replace the lecture of a cancelled class session. All instructional materials (e.g., readings, lecture notes, slides, handouts, homework assignments, lab manuals, source codes, etc.) will be distributed to students either online through Canvas or in the subsequent class session.
- If a homework assignment was due on a day on which a class session is cancelled, students are asked to upload a legible, high-quality scanned copy of their own homework solutions in PDF format on Canvas by midnight on the same day.
- If a quiz or exam was scheduled to be held on a day on which a class session is cancelled, the quiz or exam will be either postponed to the next regularly scheduled class session or held online through Canvas and Respondus Lockdown Browser and Monitor on a later day, giving students enough time to prepare the necessary software and hardware to be able to take the quiz or exam on the aforementioned online platforms.

RECORDING OF CLASS SESSIONS: Video/audio of class sessions (face-to-face or online synchronous or asynchronous) may be recorded for the benefit of students in the class. Recordings will be shared via platforms with access limited to other members of the class. The instructor will attain consent from students if recordings of student comments or images are shared with a broader audience.

COURSE COPYRIGHT: All course materials that students receive or to which students have online access are protected by copyright laws. Students may use course materials and make copies for their own use as needed, but unauthorized distribution and/or uploading of materials without the instructor's express permission is strictly prohibited. Students who engage in the unauthorized distribution of copyrighted materials may be held in violation of the Student Code of Conduct, and/or be liable under Federal and State laws. In addition, distributing completed essays, labs reports, homework, quizzes, exams, reports, or other assignments created for this course constitutes a violation of the Student Code of Conduct policy.

USE OF CANVAS: In this course we will be using Canvas to distribute and collect lecture notes, handouts, assignments, labs, quizzes, exams, activities, and/or discussions. Students should have regular access to

Canvas and their Northeastern email. Downloading the Canvas mobile app will also allow you to view content and participate in courses on an iOS or Android mobile device.

IMPORTANT ANNOUNCEMENTS: Important messages will be communicated through Canvas and/or emailed to students' Northeastern email addresses.

COURSE EVALUATION: Your feedback about the course and instructor is the only way instructors and academic units can improve the quality of a course and its content. Courses administered by the Department of Electrical and Computer Engineering are evaluated electronically via the Online Course Evaluation System. Students will receive all necessary information via email at the end of the semester. The evaluations are entirely confidential and will preserve your anonymity.

LEARNING ENVIRONMENT: Universities provide a safe haven for multiple perspectives and for disagreement and dissent. However, all of our conversations should be pursued in the spirit of mutual respect and civility. Together, we will work to create an environment in which every voice and perspective is heard and respected. The use of harmful or exclusionary language, including language that is racist, sexist, homophobic, or transphobic, would erode what we are trying to accomplish in our course and is not acceptable in the University classrooms.

UNIVERSAL DESIGN FOR LEARNING: I am committed to the principle of universal learning. This means that our classroom, virtual spaces, practices, and interactions have been designed to be as inclusive as possible. If you have a particular need, please email me or arrange a meeting with me so I can help you learn in this course. I will treat any information that you share as private and confidential. Contact the Disability Resource Center to seek official accommodations due to a disability, pregnancy, or emergency medical condition.

RELIGIOUS OBSERVANCES STATEMENT: It is my personal goal to respect the faith and religious obligations of each student. Students with exams and classes that conflict with their religious observances should notify their instructor at the beginning of the semester in order to work out a mutually agreeable alternative. Please note that, regardless of whether an absence is "excused" or "unexcused," the student is responsible for all missed course content and activities.

SYLLABUS SUBJECT-TO-CHANGE STATEMENT: This syllabus is a guide for the course. Students will be notified of any changes made by the instructor at his sole discretion with the advice of the students.